

Hundred Days of Coding in C Programing

Day-15

Name – Arayana Khare

Batch – 15

Sap id – 590042432

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases. Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

Question 1

Check whether all digits are even or not.

Step – 1 Algorithm

1. Start the program.
2. Display the menu with the options: (1) Check whether all digits are even and (2) Exit.
3. Read the user's choice.
4. Use switch-case to perform the selected operation.
5. If the choice is 1, input a positive integer.
6. Assume initially that all digits are even.
7. Extract the last digit using $\text{digit} = \text{number} \% 10$.
8. Check whether the extracted digit is odd using $\text{digit} \% 2 \neq 0$.
9. If any digit is odd, mark the result as false and stop checking.
10. Remove the last digit using $\text{number} = \text{number} / 10$ and continue until all digits are checked.
11. Display 'All digits are even' if no odd digit is found; otherwise display 'All digits not even'.
12. If the choice is 2, terminate the program.
13. For any other choice, display 'Invalid choice'.
14. Stop.

Step -2 Test Case Table

Test Case	Menu Choice	Input Number	Expected Output	Actual Output
1	1	12345	All digits not even	All digits not even
2	1	24680	All digits are even	All digits are even
3	1	2486	All digits are even	All digits are even
4	1	1234	All digits not even	All digits not even

Step – 3 Write Code in C Program

```
#include <stdio.h>

int main()
{
    int choice, num, digit, allEven, temp;

    do
    {
        printf("\n--- MENU ---\n");
        printf("1. Check whether all digits are even\n");
        printf("2. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                printf("Enter a positive integer: ");
                scanf("%d", &num);

                allEven = 1;
                temp = num;

                while (temp > 0)
                {
                    digit = temp % 10;

                    if (digit % 2 != 0)
                    {
                        allEven = 0;
                        break;
                    }

                    temp = temp / 10;
                }
            }
        }
    }
```

```
        if (allEven)
            printf("All digits are even.\n");
        else
            printf("All digits not even.\n");

        break;

    case 2:
        printf("Program ended.\n");
        break;

    default:
        printf("Invalid choice. Please try again.\n");
    }

} while (choice != 2);

return 0;
}
```

Step – 4 Compile and Execute

Case: 1

```
23
24         while (temp > 0)
25         {
26             digit = temp % 10;
27
28             if (digit % 2 != 0)
29             {
30                 allEven = 0;
31                 break;
32             }
33
34             temp = temp / 10;
35         }
36
37         if (allEven)
38             printf("All digits are even.\n");
39         else
40             printf("All digits not even.\n");
41
42         break;
43
44         case 2:
45             printf("Program ended.\n");
46             break;
47
48         default:
49             printf("Invalid choice. Please try again.\n");
50     }
51
52     } while (choice != 2);
53
54     return 0;
55 }
56
```

input

```
--- MENU ---
1. Check whether all digits are even
2. Exit
Enter your choice: 1
Enter a positive integer: 12345
All digits not even.
```

Case: 2

```
21     allEven = 1;
22     temp = num;
23
24     while (temp > 0)
25     {
26         digit = temp % 10;
27
28         if (digit % 2 != 0)
29         {
30             allEven = 0;
31             break;
32         }
33
34         temp = temp / 10;
35     }
36
37     if (allEven)
38         printf("All digits are even.\n");
39     else
40         printf("All digits not even.\n");
41
42     break;
43
44     case 2:
45         printf("Program ended.\n");
46         break;
47
48     default:
49         printf("Invalid choice. Please try again.\n");
50 }
51
52 } while (choice != 2);
53
54 return 0;
55 }
```

--- MENU ---
1. Check whether all digits are even
2. Exit
Enter your choice: 1
Enter a positive integer: 24680
All digits are even.

Case: 3

```
21     allEven = 1;
22     temp = num;
23
24     while (temp > 0)
25     {
26         digit = temp % 10;
27
28         if (digit % 2 != 0)
29         {
30             allEven = 0;
31             break;
32         }
33
34         temp = temp / 10;
35     }
36
37     if (allEven)
38         printf("All digits are even.\n");
39     else
40         printf("All digits not even.\n");
41
42     break;
43
44     case 2:
45         printf("Program ended.\n");
46         break;
47
48     default:
49         printf("Invalid choice. Please try again.\n");
50 }
51
52 } while (choice != 2);
53
54 return 0;
55 }
```

--- MENU ---

1. Check whether all digits are even

2. Exit

Enter your choice: 1

Enter a positive integer: 2486

All digits are even.

Case: 4

```
21     allEven = 1;
22     temp = num;
23
24     while (temp > 0)
25     {
26         digit = temp % 10;
27
28         if (digit % 2 != 0)
29         {
30             allEven = 0;
31             break;
32         }
33
34         temp = temp / 10;
35     }
36
37     if (allEven)
38         printf("All digits are even.\n");
39     else
40         printf("All digits not even.\n");
41
42     break;
43
44     case 2:
45         printf("Program ended.\n");
46         break;
47
48     default:
49         printf("Invalid choice. Please try again.\n");
50 }
51
52 } while (choice != 2);
53
54 return 0;
55 }
```

--- MENU ---
1. Check whether all digits are even
2. Exit
Enter your choice: 1
Enter a positive integer: 1234
All digits not even.

Step – 5 Conclusion

The menu-driven C program was successfully implemented using switch-case. It correctly checks each digit of a positive integer and determines whether all digits are even. The program was compiled and tested with multiple inputs, and all test cases produced the expected results.

Question 2

Check whether all digits are prime or not

Step – 1 Algorithm

1. Start the program.
2. Display the menu with the options: Check whether all digits are prime and Exit.
3. Read the user's choice.
4. Use switch-case to perform the selected operation.
5. If the choice is 1, input a positive integer.
6. Assume initially that all digits are prime.
7. Extract the last digit using $\text{digit} = \text{number} \% 10$.
8. Check whether the digit is one of the prime digits 2, 3, 5, or 7.
9. If any digit is not 2, 3, 5, or 7, mark the result as false and stop checking.
10. Remove the last digit using $\text{number} = \text{number} / 10$ and continue until all digits are checked.
11. Display 'All digits are prime' if every digit is prime; otherwise display 'All digits are not prime'.
12. If the choice is 2, terminate the program.
13. For any other choice, display 'Invalid choice'.
14. Stop.

Step -2 Test Case Table

Test Case	Menu Choice	Input Number	Expected Output	Actual Output
1	1	2357	All digits are prime	All digits are prime
2	1	2379	All digits are not prime	All digits are not prime
3	1	222	All digits are prime	All digits are prime
4	1	1234	All digits are not prime	All digits are

Step – 3 Write Code in C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int choice, num, digit, allPrime, temp;
```

```
do
```

```
{
```

```
printf("\n--- MENU ---\n");
```

```
printf("1. Check whether all digits are prime\n");
```

```
printf("2. Exit\n");
```

```
printf("Enter your choice: ");
```

```
scanf("%d", &choice);
```

```
switch (choice)
```

```
{
```

```
case 1:
```

```
printf("Enter a positive integer: ");
```

```
scanf("%d", &num);
```

```
allPrime = 1;
```

```
temp = num;
```

```
while (temp > 0)
```

```
{
```

```
digit = temp % 10;

if (digit != 2 && digit != 3 &&
    digit != 5 && digit != 7)
{
    allPrime = 0;
    break;
}

temp = temp / 10;
}

if (allPrime)
    printf("All digits are prime.\n");
else
    printf("All digits are not prime.\n");

break;

case 2:
    printf("Program ended.\n");
    break;

default:
    printf("Invalid choice. Please try again.\n");
}

} while (choice != 2);

return 0;
}
```

Step – 4 Compile and Execute

Case: 1

```
17      case 1:
18          printf("Enter a positive integer: ");
19          scanf("%d", &num);
20
21          allPrime = 1;
22          temp = num;
23
24          while (temp > 0)
25          {
26              digit = temp % 10;
27
28              if (digit != 2 && digit != 3 &&
29                  digit != 5 && digit != 7)
30              {
31                  allPrime = 0;
32                  break;
33              }
34
35              temp = temp / 10;
36          }
37
38          if (allPrime)
39              printf("All digits are prime.\n");
40          else
41              printf("All digits are not prime.\n");
42
43          break;
44
45      case 2:
46          printf("Program ended.\n");
47          break;
48
49      default:
50          printf("Invalid choice. Please try again.\n");

```

input

```
--- MENU ---
1. Check whether all digits are prime
2. Exit
Enter your choice: 1
Enter a positive integer: 2357
All digits are prime.
```

Case: 2

```
22     temp = num;
23
24     while (temp > 0)
25     {
26         digit = temp % 10;
27
28         if (digit != 2 && digit != 3 &&
29             digit != 5 && digit != 7)
30         {
31             allPrime = 0;
32             break;
33         }
34
35         temp = temp / 10;
36     }
37
38     if (allPrime)
39         printf("All digits are prime.\n");
40     else
41         printf("All digits are not prime.\n");
42
43     break;
44
45     case 2:
46         printf("Program ended.\n");
47         break;
48
49     default:
50         printf("Invalid choice. Please try again.\n");
51 }
52
53 } while (choice != 2);
54
55 return 0;
56 }
57
```

input

```
--- MENU ---
1. Check whether all digits are prime
2. Exit
Enter your choice: 1
Enter a positive integer: 2379
All digits are not prime.
```

Case: 3

```
22         temp = num;
23
24         while (temp > 0)
25         {
26             digit = temp % 10;
27
28             if (digit != 2 && digit != 3 &&
29                 digit != 5 && digit != 7)
30             {
31                 allPrime = 0;
32                 break;
33             }
34
35             temp = temp / 10;
36         }
37
38         if (allPrime)
39             printf("All digits are prime.\n");
40         else
41             printf("All digits are not prime.\n");
42
43         break;
44
45         case 2:
46             printf("Program ended.\n");
47             break;
48
49         default:
50             printf("Invalid choice. Please try again.\n");
51     }
52
53     } while (choice != 2);
54
55     return 0;
56 }
57
```

```
--- MENU ---
1. Check whether all digits are prime
2. Exit
Enter your choice: 1
Enter a positive integer: 222
All digits are prime.
```

Case: 4

```
22         temp = num;
23
24         while (temp > 0)
25         {
26             digit = temp % 10;
27
28             if (digit != 2 && digit != 3 &&
29                 digit != 5 && digit != 7)
30             {
31                 allPrime = 0;
32                 break;
33             }
34
35             temp = temp / 10;
36         }
37
38         if (allPrime)
39             printf("All digits are prime.\n");
40         else
41             printf("All digits are not prime.\n");
42
43         break;
44
45         case 2:
46             printf("Program ended.\n");
47             break;
48
49         default:
50             printf("Invalid choice. Please try again.\n");
51     }
52 } while (choice != 2);
53
54 return 0;
55 }
56
57
```

inp

```
--- MENU ---
1. Check whether all digits are prime
2. Exit
Enter your choice: 1
Enter a positive integer: 1234
All digits are not prime.
```

Step – 5 Conclusion

The menu-driven C program was successfully implemented using switch-case. It checks every digit of a positive integer and correctly determines whether all the digits are prime. The program was tested with different inputs and produced the expected results.